

Zepeng Zhang

Intelligent Maintenance and Operations Systems (IMOS) Lab
École Polytechnique Fédérale de Lausanne (EPFL)

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EDUCATION

- Ph.D. Electrical Engineering, École Polytechnique Fédérale de Lausanne, 2023.09-present
Supervisor: Prof. Olga Fink
- M.S. Information and Communication Engineering, ShanghaiTech University, 2020.09-2023.07
Overall GPA: 3.95/4.00; Major GPA: 4.00/4.00;
- B.Eng. Measuring & Control Technology and Instrumentations, Wuhan University, 2016.09-2020.05

EXPERIENCES

- 2025.04–2025.08 Visiting Student, [Télécom Paris, Institut Polytechnique de Paris](#)
- 2023.02–2023.04 Research Intern, [Peking University](#)
- 2019.12–2020.09 Visiting Student, [ShanghaiTech University](#)
- 2019.07–2019.09 Visiting Student, [City University of HongKong](#)
- 2018.01–2020.06 Member, [WHUAI \(ZhuYing Team\)](#)

CURRENT RESEARCH INTERESTS

Graph Machine Learning, Graph Signal Processing, Generative Model

PUBLICATIONS

- Domain Adaptive Unfolded Graph Neural Networks
Zepeng Zhang and Olga Fink
AAAI Conference on Artificial Intelligence (AAAI), 2025. [[paper](#)]
- Algorithm-Informed Graph Neural Networks for Leakage Detection and Localization in Water Distribution Networks
Zepeng Zhang and Olga Fink
Reliability Engineering & System Safety (RESS), 2025. [[paper](#)]
- RINS-T: Robust Implicit Neural Solvers for Time Series Linear Inverse Problems
Keivan Faghieh Niresi, **Zepeng Zhang** and Olga Fink
IEEE Transactions on Instrumentation and Measurement (TIM), 2025.
- GeoLink: Empowering Remote Sensing Foundation Model with OpenStreetMap Data
Lubin Bai, Xiuyuan Zhang, Siqi Zhang, **Zepeng Zhang**, Haoyu Wang, Wei Qin, and Shihong Du
In The Thirty-ninth Annual Conference on Neural Information Processing Systems (NeurIPS), 2025. [[paper](#)]

5. Weighted Sum-Rate Maximization using MM: Convergence Rate and Deep Adaptation
Zhexiong Yang, **Zepeng Zhang**, and Ziping Zhao
In European Signal Processing Conference (EUSIPCO), 2025.
6. Graph Neural Networks With Adaptive Structures
Zepeng Zhang, Songtao Lu, Zengfeng Huang, and Ziping Zhao
IEEE Journal of Selected Topics in Signal Processing: Special Series on AI in Signal & Data Science (JSTSP), 2024. [[paper](#)]
7. Accelerating quadratic transform and WMMSE
Kaiming Shen, Ziping Zhao, Yannan Chen, **Zepeng Zhang**, and Hei Victor Cheng
IEEE Journal on Selected Areas in Communications: Special Issue on Advanced Optimization Theory and Algorithms for Next Generation Wireless Communication Networks (JSAC), 2024. [[paper](#)]
8. Graph Neural Networks With Adaptive Structures
Zepeng Zhang, Songtao Lu, Zengfeng Huang, and Ziping Zhao
In The Graph Signal Processing Workshop (GSP), 2024.
9. A Joint Graph Signal and Laplacian Denoising Network
Zepeng Zhang and Ziping Zhao
The Asia-Pacific Signal and Information Processing Conference (ACSSC/Asilomar), 2024.
10. Accelerating quadratic transform and WMMSE
Kaiming Shen, Ziping Zhao, Yannan Chen, **Zepeng Zhang**, and Hei Victor Cheng
The IEEE International Symposium on Information Theory (ISIT), 2024. [[paper](#)]
11. Enhancing the Efficiency of WMMSE and FP for Beamforming by Minorization-Maximization
Zepeng Zhang, Ziping Zhao, and Kaiming Shen
IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2023 [[paper](#)]
12. Weighted Sum-Rate Maximization for Multi-Hop RIS-Aided Multi-User Communications: A Minorization-Maximization Approach.
Zepeng Zhang and Ziping Zhao
IEEE International Workshop on Signal Processing Advances in Wireless Communications (SPAWC) 2021. [Best Student Paper Award Finalist](#) [[paper](#)]
13. Vast Portfolio Selection With Submodular Norm Regularizations.
Zepeng Zhang, and Ziping Zhao
European Signal Processing Conference (EUSIPCO), 2021. [[paper](#)]
14. Multi-Period Portfolio Optimization for Financial Index Tracking.
Xiuyuan Huang, **Zepeng Zhang**, and Ziping Zhao
Annual Asilomar Conference on Signals, Systems, and Computers (ACSSC/Asilomar) 2021. [[paper](#)]
15. Scalable Financial Index Tracking With Graph Neural Networks.
Zepeng Zhang, and Ziping Zhao
IEEE Statistical Signal Processing Workshop (SSP) 2021. [[paper](#)]
16. A Deep Learning-Aided Approach to Portfolio Design for Financial Index Tracking.
Zepeng Zhang, and Ziping Zhao
Annual Asilomar Conference on Signals, Systems, and Computers (ACSSC/Asilomar) 2020. [[paper](#)]

PREPRINT

1. Discerning and enhancing the weighted sum-rate maximization algorithms in communications
Zepeng Zhang, Ziping Zhao, Kaiming Shen, Daniel P Palomar, and Wei Yu
arXiv preprint arXiv:2311.04546, 2023. [[paper](#)]
2. Towards understanding graph neural networks: An algorithm unrolling perspective
Zepeng Zhang and Ziping Zhao
arXiv preprint arXiv:2206.04471, 2022. [[paper](#)]
3. Rate maximizations for reconfigurable intelligent surface-aided wireless networks: A unified framework via block minorization-maximization
Zepeng Zhang and Ziping Zhao
arXiv preprint arXiv:2105.02395, 2021. [[paper](#)]

AWARDS AND HONORS

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|------|---|
| 2021 | Best Student Paper Award Finalist in IEEE International Workshop on Signal Processing Advances in Wireless Communications (SPAWC) |
| 2019 | Yu Gang Scholarship (Top 1%, Rank 1st out of the department), WHU
Outstanding Student First-Class Scholarship (Top 5%), WHU |
| 2018 | SAMSUNG Scholarship (Top 1%, Rank 1st out of the department), WHU
Outstanding Student First-Class Scholarship (Top 5%), WHU
1 st prize of RoboCup China Open
1 st prize of Asia and Pacific Mathematical Contest in Modeling
3 rd prize of The Chinese Robot Skills Competitions |
| 2017 | Outstanding Student Second-Class Scholarship (Top 15%), WHU
1 st prize of The Chinese Mathematics Competitions, Hubei Prov. |

TEACHING ASSISTANT

EPFL

[CIVIL 426](#): Machine learning for predictive maintenance applications 2023-24 Fall, 2024-25 Fall, 2025-26 Fall

[CIVIL 332](#): Data Science for infrastructure condition monitoring 2023-24 Spring, 2024-25 Spring

[MATH 106](#): Math Analyse II, 2024-25 Spring

ShanghaiTech

[CS182](#): Introduction to Machine Learning 2021-22 Fall

[SI 231](#): Matrix Analysis and Computations 2020-21 Spring, 2021-22 Spring

ACADEMIC SERVICE

Journal Reviewer: IEEE Transactions on Wireless Communications, Mechanical Systems and Signal Processing, IEEE Open Journal of Signal Processing, Neurocomputing

Conference Reviewer: ICML, NeurIPS, LOG, AISTATS, WWW, ICASSP, ECAI, MLSP

SKILLS

Python, MATLAB, CVX, PyTorch, PyG, LaTeX

Toefl: 105 (Reading: 30; Listening: 27; Speaking: 22; Writing: 26)